

Build Options Reference: Every Tweak, Organized by Option

`docs/tweak-catalog.md` and `docs/script-comparison.md` organize this repo's tweaks **by source tool** ("this repo vs. ntdevlabs vs. tiny11-automated vs. winutil"). This document organizes the *same* tweaks a different way: **by the** `asl-win11maker.ps1` / `asl-win11-multibuild.ps1` **option that turns them on**, so you can look at one CLI flag (or one multi-build profile) and see everything it actually does, when it happens, and where the implementation came from. Every row here is a pointer, not a re-description — full detail lives in `docs/tweak-catalog.md` (packages/tasks/registry), `CLAUDE.md` (parameters/architecture), or `docs/answer-file-generators-options.md` / `docs/rufus-feature-options.md` (answer-file technique sourcing).

Current as of the 2026-10-05 pipeline (`asl-win11maker.ps1` lines 190–403); re-check this doc's line references if the pipeline is reordered.

Taxonomy

Three independent dimensions, used consistently across every table below:

Category — what kind of thing is being changed:

Category	Examples
Packages	Provisioned appx/MSIX removal, Core-only CBS/FoD system packages
Scheduled Tasks	Telemetry/CEIP/WU task-definition file deletion
Registry & Policy	<code>Disable-*</code> / <code>Enable-*</code> / <code>Set-*</code> offline hive edits
Services	Service <code>Start</code> value changes (folded into several <code>Disable-*</code> functions, not a separate category in code)
Files & Components	WinSxS/Defender/WinRE/Edge-WebView file-level removal, <code>.NET 3.5</code> enablement
Drivers	<code>install.wim</code> driver injection, <code>\$WinpeDriver\$</code> staging, <code>boot.wim</code>
Answer-File & OOBE Behavior	Product key, locale, local account, hardware-bypass <code>reg add</code> commands, edition pinning

Phase — *when* the change takes effect, independent of category:

Phase	What it means	Who in this repo's ecosystem uses it
Offline pre-boot	DISM mounts <code>install.wim</code> / <code>boot.wim</code> , the pipeline edits the mounted files/hives directly, DISM commits and unmounts — done before the ISO is even built	This repo's default approach for nearly everything (packages, tasks, most registry tweaks, drivers, file/component removal)
Setup-time declarative	A plain <code><Component></code> value in <code>autounattend.xml</code> that Windows Setup itself reads and applies — no script runs	This repo's product key, locale (<code>Add-AnswerFileLocale</code>), local account (<code>Add-AnswerFileLocalAccount</code>), edition index injection
Setup-time embedded script	A PowerShell (or <code>reg add</code>) command embedded in <code>autounattend.xml</code> via <code>RunSynchronousCommand</code> / <code>RunAsynchronousCommand</code> , executed live during a Setup pass (windowsPE, specialize, or oobeSystem)	This repo's <code>-BypassMode Unattend / Both</code> (<code>Add-AnswerFileBypassCommands</code>); <code>UnattendedWinstall</code> and <code>unattend-generator</code> use this for <i>everything</i> including package removal — see <code>docs/answer-file-generators-options.md</code>
Post-OOBE first-logon	Needs an actual logged-on user/session context, so it can only run after OOBE finishes — a scheduled task or <code>RunOnce</code> entry	This repo's <code>Install-VirtioGuestToolsAtFirstLogon</code>

Mechanism — *how* it's technically done: DISM API (`Invoke-Dism` / `dism.exe`) · offline registry hive mount (`Mount-OfflineRegistryHives` + `Set-RegistryValue`) · raw file ops (`Remove-Item` / `takeown` / `icacls`) · answer-file XML component · answer-file embedded script · scheduled task definition file.

Default serviceable build (no switches)

Always-on `REQUIRED` / active- `OPTIONAL` tweaks from `asl-win11maker.ps1` lines 336–392 with no switches set.

Item	Category	Phase	Mechanism	Source
Provisioned app package removal (49 active of 63 prefixes)	Packages	Offline pre-boot	DISM Remove-AppxProvisionedPackage	docs/tweak-catalog.md §1
Remove-OneDriveSetup	Files & Components	Offline pre-boot	Raw file ops (takeown / icacls + delete)	docs/script-comparison.md §1 (OneDrive removal row); ntdevlabs technique
Remove-IsoSupportFolder	Files & Components	Offline pre-boot	Raw file ops	winutil (docs/script-comparison.md §4 item 4)
Invoke-HardwareBypassStrategy	Answer-File & OOBE Behavior	(no-op — default -BypassMode None)	—	—
Disable-SponsoredApps	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs — docs/tweak-catalog.md §3
Enable-LocalAccountOOBE	Answer-File & OOBE Behavior	Setup-time embedded script (BypassNRO)	Offline hive mount for the image-side half + answer-file reg add	Rufus UNATTEND_NO_ONLINE_ACCOUNT — docs/rufus-feature-options.md
Disable-ReservedStorage	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-BitLockerAutoEncryption	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs (Rufus also sets TCGSecurityActivationDisabled , not yet ported — docs/rufus-feature-options.md §4)
Disable-Telemetry	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-ChatIcon	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-OneDriveSync	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-DevHomeOutlookInstall	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-Copilot	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-TeamsInstall	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Disable-NewOutlook	Registry & Policy	Offline pre-boot	Offline hive mount	ntdevlabs
Scheduled task removal (5 active of 11 paths)	Scheduled Tasks	Offline pre-boot	Delete task definition .xml files	docs/tweak-catalog.md §2
Product key / GVLK / edition index	Answer-File & OOBE Behavior	Setup-time declarative	Answer-file XML (ConvertTo-Tiny11AnswerFile)	winutil's ConvertTo-WinUtilISOAnswerFile pattern, extended — docs/script-comparison.md §4 item 2
Locale injection (Add-AnswerFileLocale)	Answer-File & OOBE Behavior	Setup-time declarative	Answer-file XML	Rufus UNATTEND_DUPLICATE_LOCALE -style — docs/rufus-feature-options.md §1
Set-Tiny11EditionConfig (ei.cfg/PID.txt)	Answer-File & OOBE Behavior	Offline pre-boot (writes to ISO content tree)	Raw file ops	winutil Write-WinUtilISOEditionConfig — docs/script-comparison.md §4 item 3

Commented out by default (safe to uncomment, no code change needed): Microsoft.MPEG2VideoExtension , Microsoft.Recall / Microsoft.Windows.Recall , Microsoft.ScreenSketch , Microsoft.StorePurchaseApp , Microsoft.WebMediaExtensions , Microsoft.Windows.AI / AIFabric / CoreAI , Microsoft.Windows.Photos , MicrosoftWindows.Client.WebExperience (packages — docs/tweak-catalog.md §1); the 6 winutil WU-task paths (scheduled tasks — §2); Disable-WindowsUpdate , Disable-DiagnosticServices , Disable-WindowsAI (registry — §3).

- Core

Everything in the default build above, **plus**:

Item	Category	Phase	Mechanism	Source
Remove-SystemPackages (CBS/FoD packages)	Packages	Offline pre-boot	DISM /Remove-Package	asl-win11-coremaker.ps1 lineage, folded into -Core — CLAUDE.md
Enable-DotNet35 (if confirmed)	Files & Components	Offline pre-boot	DISM /Enable-Feature from source media	asl-win11-coremaker.ps1 lineage
Remove-EdgeWebViewWinSxS	Files & Components	Offline pre-boot	DISM WinSxS package removal	tiny11-automated (WinSxS Edge-WebView removal — docs/script-comparison.md §1 Edge row)
Remove-WindowsRecoveryEnvironment	Files & Components	Offline pre-boot	Raw file ops + BCD edits	asl-win11-coremaker.ps1 lineage — see tiny11-automated's 8bde4bd WinRE staging fix, docs/script-comparison.md §4 item 9, unevaluated against this function
Compress-WinSxS	Files & Components	Offline pre-boot	DISM /StartComponentCleanup variant	asl-win11-coremaker.ps1 lineage
Remove-Edge	Files & Components	Offline pre-boot	Raw file ops + registry	ntdevlabs/tiny11-automated — docs/script-comparison.md §1 Edge row
Disable-WindowsUpdate	Registry & Policy	Offline pre-boot	Offline hive mount	Merged asl-win11-coremaker.ps1 + winutil — docs/tweak-catalog.md §3
Disable-WindowsDefender	Registry & Policy	Offline pre-boot	Offline hive mount	asl-win11-coremaker.ps1 lineage
Set-BootImageSetupCmdLine	Registry & Policy (boot.wim)	Offline pre-boot	Offline hive mount on boot.wim	asl-win11-coremaker.ps1 lineage
Export to install.esd instead of install.wim	— (export format)	Offline pre-boot	DISM /Export-Image	asl-win11-coremaker.ps1 lineage

Trade-off this option exists to document: **non-serviceable** — cannot take Windows Update, add features/languages, or use Windows Recovery afterward. Disposable/testbed images only.

-BypassMode <None|Hive|Unattend|Both>

Hardware-check bypass is the one axis implemented as **alternative mechanisms for the same 5 keys**, not alternative tweaks:

Mode	Phase	Mechanism	Source
None (default)	—	No-op. Setup enforces TPM/Secure Boot/RAM checks normally.	Deliberate behavior change from pre-combined-builder scripts — CLAUDE.md
Hive	Offline pre-boot	Set-BypassHardwareChecks edits install.wim + boot.wim offline hives directly	ntdevlabs/tiny11-automated (identical key set)
Unattend	Setup-time embedded script	Add-AnswerFileBypassCommands emits reg add as RunSynchronousCommand in autounattend.xml	Rufus wue.c UNATTEND_SECUREBOOT_TPM_MINRAM — docs/rufus-feature-options.md §1. Note: unattend-generator's BypassModifier (modifier/Bypass.cs) does the same 3-key subset (BypassTPMCheck / BypassSecureBootCheck / BypassRAMCheck) in the windowsPE pass — confirms the technique independently, but doesn't cover this repo's extra 2 keys (AllowUpgradesWithUnsupportedTPMOrCPU + notification-cache suppression).
Both	Both of the above	Both	—

The 5 keys themselves: BypassTPMCheck, BypassSecureBootCheck, BypassRAMCheck, BypassStorageCheck, BypassCPUCheck (all HKLM\SYSTEM\Setup\LabConfig) plus AllowUpgradesWithUnsupportedTPMOrCPU and notification-cache suppression — docs/tweak-catalog.md §3.

This is also the **hardware compatibility** axis referenced in the settings-strategy doc, together with driver injection below.

-KeepCorporateApps

Item	Category
Carves Microsoft.Copilot / Microsoft.Windows.Copilot / Microsoft.Windows.Teams / MSTeams / MicrosoftTeams / Microsoft.OutlookForWindows out of the removal list	Packages
Skips Remove-OneDriveSetup	Files & Components
Skips Disable-ChatIcon / Disable-OneDriveSync / Disable-DevHomeOutlookInstall / Disable-Copilot / Disable-TeamsInstall / Disable-NewOutlook	Registry & Policy
Skips Enable-LocalAccountOOBE (even if -LocalAccountName given — warns, clears the var)	Answer-File & OOBE Behavior

Every other debloat/telemetry tweak in the default build still applies. Intended for Entra ID / Autopilot / Intune-enrolled images — this is the **account mode = Online** axis.

-LocalAccountName <name>

Item	Category	Phase	Mechanism	Source
Add-AnswerFileLocalAccount	Answer-File & OOBE Behavior	Setup-time declarative (<UserAccounts> <LocalAccounts>)	Answer-file XML	Rufus UNATTEND_SET_USER — docs/rufus-feature-options.md ("Different from Rufus" section documents this repo's plaintext-password choice vs. Rufus's blank-password + forced-change)

This is the **account mode = Local** axis. Security footnote carried over from CLAUDE.md : the password equals the account name in plaintext. Per docs/answer-file-generators-options.md 's sensitive-file finding, that plaintext (plus -ProductKey , if given) used to be left readable in C:\Windows\Panther\unattend.xml and Windows\System32\Sysprep\autounattend.xml on the installed system with no cleanup — **now fixed** via Remove-SensitiveAnswerFilesAtFirstLogon , a RunOnce entry (staged whenever Enable-LocalAccountOOBE runs, i.e. every build except -KeepCorporateApps) that deletes all three known copies at first logon. See docs/optimization-checklists.md .

-InjectSystemDrivers / -DriverPath <folder> / -InjectVirtioDrivers [-VirtioIso <path>]

Item	Category	Phase	Mechanism	Source
Export-HostSystemDrivers (host drivers only)	Drivers	Offline pre-boot (host export happens at build time)	Export-WindowsDriver -Online on the build machine	winutil -InjectCurrentSystemDrivers — docs/script-comparison.md §4 item 7
Add-DriversToImage (install.wim, full driver set)	Drivers	Offline pre-boot	DISM /Add-Driver /Recurse	winutil, same item
Add-WinPEStorageDrivers (storage-class subset → \$WinpeDriver\$ at ISO root)	Drivers	Offline pre-boot (ISO content staging; Setup loads it live during its own windowsPE pass)	Raw file copy, auto-loaded by Setup	Ported from winutil's Add-WinUtilISOStagedDrivers , confirmed via KB2686316 — CLAUDE.md
Resolve-VirtioDriverSource / Add-VirtioDriversToImage	Drivers	Offline pre-boot	DISM /Add-Driver , 7-Zip extraction instead of mount	This repo, went beyond winutil's scope per docs/script-comparison.md §4 item 7
Install-VirtioGuestToolsAtFirstLogon	Drivers (guest tools)	Post-OOBE first-logon	Scheduled task / first-logon script	This repo

Remediated (finding from the 2026-10-05 submodule refresh — see docs/script-comparison.md 's candidate #10 for the original writeup): winutil's abcbc23 fix ("inject Setup storage into boot.wim") found that its filename-based storage-driver heuristic (iaahci|iastor|vmd|irst|rst) let non-storage "RST companion" INFs slip into \$WinpeDriver\$ as false positives, and that \$WinpeDriver\$ staging alone wasn't a sufficient delivery path — it now *also* mounts boot.wim index 2 and injects the SCSIAdapter/HDC driver packages directly via DISM. Both halves of that fix are now ported here:

- Test-StorageDriverInf no longer has a filename fallback (this repo's version additionally covered viostor|vioscsi|nvme for virtio/NVMe) — it matches only the INF's own Class=SCSIAdapter|HDC directive, same as winutil's narrowed check.
- Update-BootImage now re-injects the already-staged \$WinpeDriver\$ folder into boot.wim directly via Add-DriversToImage (DISM /Add-Driver /Recurse) right after mounting, reusing the exact already-storage-filtered package set rather than bloating boot.wim with a full

driver-source rescan.

\$WinpeDriver\$ staging (via KB2686316 auto-load) remains the primary mechanism; the boot.wim injection is a second delivery path for the same packages. Still worth a real-hardware/VM boot test after relying on `-InjectSystemDrivers / -DriverPath / -InjectVirtioDrivers` for an unusual storage controller — this closes the known false-positive-matching gap, it doesn't guarantee every possible controller's INF is well-formed.

This is the other half of the **hardware compatibility** axis (alongside `-BypassMode`).

asl-win11-multibuild.ps1 profiles

Each profile is a different combination of the options above, run as a separate `asl-win11maker.ps1` child process:

Profile	<code>-BypassMode</code>	<code>-InjectSystemDrivers</code>	Everything else
Standard	Forced <code>None</code> regardless of orchestrator input	No	Default build, as above
Entra	Whatever the orchestrator was given (default <code>None</code>)	Yes	Default build + <code>-KeepCorporateApps</code> effects above; ignores <code>-LocalAccountName</code>
Optimized	Whatever was given	No	Whatever <code>-LocalAccountName</code> / <code>-ProductKey</code> / driver switches were given, used as-is

All three always run with `-UseSourceCache` under the hood so only the first profile touches the mounted ISO. See `CLAUDE.md`'s "Multi-profile builds" section for the full parameter-forwarding details.

See also

- `docs/tweak-catalog.md` — per-tweak four-way source comparison (packages/tasks/registry detail).
- `docs/script-comparison.md` — tool-level architecture comparison, plus the two new candidates from the 2026-10-05 submodule refresh.
- `docs/answer-file-generators-options.md` — the phase taxonomy's "Setup-time embedded script" technique, sourced from `UnattendedWininstall / unattend-generator`.
- `docs/rufus-feature-options.md` — Rufus-sourced answer-file techniques.
- `docs/settings-config-strategy.md` — how this option-first view maps onto a proposed config-file/profile system (package aggressiveness, account mode, hardware compatibility as three independent axes).
- `docs/optimization-checklists.md` — decision checklists built from this document.